MILITARY SPECIFICATION

BORING MACHINE, CAM AND MAIN BEARINGS

(STATIONARY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of stationary, universal, boring machine for line boring and facing cam and main bearing bores in motor vehicle engines.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807 - Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-S-12514 - Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.
AGMA 390.03 - Gear Handbook, Volume 1, Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 North Fort Myer Dr., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.8 - Safety Requirements for the Construction, Care, and Use of Drilling, Milling, and Boring Machines.
ANSI IEEE 268- Metric Practice.

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(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy MA 02269.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete boring machine for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The boring machine shall be new and of the manufacturer's current design meeting the requirements specified herein. The machine shall have a stationary base and either a vertically adjustable engine support bed or a vertically adjustable boring spindle. The bed shall be capable of supporting the engine block sizes specified in 3.5.2. The machine shall be electric motor driven and shall provide automatic cutting feed for boring and facing operations. Parts

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subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the machine or machine components.

* 3.2.1 Measurement system. The US customary system of units (US) or the International System of Units (SI) may be used in the design and construction of the machine. In this specification all measurements, dimensions, sizes, and capacities are given in the US customary system of units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI IEEE 268 and ASTM E380.

3.2.2 Measuring and indicating devices. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI system of measurements. When a dual US and SI system is furnished, feed dials shall have independent zero adjustments and shall be calibrated so that the last dial graduation progresses into, and is continuous with, the first dial graduation as the dial is rotated through the zero position. When only one system of measurement is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the machine tool industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

* 3.4 Construction. The boring machine shall be constructed of parts and materials which are without defects and are free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine, at its maximum rating and capacity, without distortion. Construction shall be such that, when installed and connected to the correct power supply, the machine shall be ready for operation.

* 3.4.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

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3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

* 3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with approved engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.4.4 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in all soldering operations.

* 3.4.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.4.6 Gears. All gears and pinion designs selected for use in the boring machine or its components shall be manufactured to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122/1 and ASTM E380-84.

* 3.4.7 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

* 3.4.8 Safety and health requirements. Covers, guards, and other safety devices shall be provided. The devices shall not interfere with operation of the boring machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of components. All parts, components, mechanisms, and assemblies of the boring machine, whether specifically required herein or not, shall comply with those requirements of OSHA 2206 and ANSI B11.8 that are applicable to boring machines. In the event of conflict between the requirements, the OSHA standards shall apply. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.4.9 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.4.10 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous material. External surfaces shall be smooth and all edges shall be rounded or beveled.

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3.5 Performance. The boring machine shall be capable of restoring the original bearing alignment to engine block bearing bores. The boring bar shall be rigidly supported, shall rotate in bearings that prevent deflection or sag, and shall operate without chatter.

3.5.1 Accuracy. The boring machine shall be capable of producing finished bearing surfaces that are concentric and without taper to ± 0.0005 inch, and that have a surface roughness no greater than 20 microinches arithmetical average (aa).

* 3.5.2 Capacity. The boring machine shall be capable of performing the operations specified herein within a bearing bore diameter range of 1 3/16 to 6 1/2 inches in engine blocks up to and including:

72 inches over-all spread of cam or main bearing bores.
16 inches from side of crankcase to main bearing bore centerline.
32 inches from table to bearing bore centerline.

3.6 Details of components. The boring machine shall consist of, but not be limited to, the principal components specified in 3.6.1 thru 3.6.5.3.

* 3.6.1 Base. The boring machine base shall be equipped with precision machined ways that shall provide for boring or facing bar adjustment.

3.6.2 Bar supports. Not less than two bar supports shall maintain the boring or facing bar alignment.

* 3.6.3 Vertical adjustment. The boring machine shall provide for vertical adjustment by means of either an adjustable engine support bed or a vertically adjustable boring spindle.

* 3.6.4 Feed control. The machine shall have an adjustable hydraulic or electric driving unit and feed control to drive and feed the boring bar.

3.6.5 Electrical equipment.

* 3.6.5.1 Motors. Unless otherwise specified (see 6.2.1), the motor shall be capacitor start, shall be of not less than 1/4 horsepower (hp), and shall operate on 115 volt, 60 hertz, single phase current. The motor shall have a continuous duty rating and shall conform to the requirements of CC-M-1807, with class B insulation. If a 3/4 hp or larger motor using three phase current is required, it shall be an induction squirrel-cage type equipped with a starter conforming to type I, class 2 requirements of MIL-S-12514.

* 3.6.5.2 Electrical components. Electrical components shall conform to, and shall be located in accordance with, NFPA 79. No electrical component shall be used in any application where the manufacturer's rating for the part does not meet or exceed the requirements.

* 3.6.5.3 Power cord. Unless otherwise specified (see 6.2.1), each boring machine shall be furnished with an electrical power supply cord not less than eight feet in length. The wire shall be no smaller than No. 14 AWG, type SO, heavy duty, conforming to NFPA 70. For single phase motors, a three conductor power cord shall be provided with a three prong polarized connector plug, either of the grip-cord type or molded-on type. A mating three to two parallel prong

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adapter with grounding lead shall also be furnished. For three phase motors, a four conductor power cord shall terminate in not less than one inch of stripped tinned wire, without a connector plug.

* 3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the standard equipment specified in 3.8.1 thru 3.8.5 shall be provided with each boring machine.

* 3.8.1 Fixtures. All set-up and operation fixtures required for the capacity or operation of the boring machine in straight line cylinder engine blocks or in V-6 or V-8 engine blocks, within the size range specified herein, shall be provided.

3.8.2 Measuring devices. All gages, special micrometers, and dial indicators required for engine block alignment or tool bit adjustment shall be provided. The measuring devices shall be permanently marked with clear, bold-face characters that can be easily read.

3.8.3 Boring and facing bars. The boring and facing bars provided shall be furnished with internal tool bit holder devices.

3.8.4 Tools. A set of all special and standard tools and wrenches required for the set-up operation of the boring machine shall be provided.

* 3.8.5 Tool bits. Not less than three sets of insert type carbide bits shall be provided. Each set shall consist of one bit ground for each boring and facing operation. The tool bits shall have a hardness of not less than 63 to 65 on the Rockwell C scale.

* 3.9 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.10 Product identification. The boring machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.11 Workmanship. Standards of workmanship shall assure that the boring machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection

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requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests under 4.7. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

* 4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.6).
(b) Functional test (see 4.7.2).
(c) Packaging inspection (see 4.8).

* 4.5 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105. For product examination (see 4.6) and the functional test (see 4.7.2) use General Inspection Level II and an Acceptable Quality Level (AQL) of 1.0 percent. For packaging inspection (see 4.8) use Special Inspection Level S-4 and an AQL of 4.0 percent.

* 4.6 Examination. Visually and manually examine the boring machine to determine conformance with the requirements of 3.2 through 3.3.1 and 3.4 through 3.4.10. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.7 Tests.

* 4.7.1 Performance test. The boring machine shall be used to refinish the cam and main bearing bores of a standard commercial six or eight cylinder engine block that is within the capacity limits specified in 3.5.2. The boring machine

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performance and the finished bore surfaces shall comply with the requirements specified in Section 3.

* 4.7.2 Functional test. Each machine shall be operated a minimum of 30 minutes to verify proper operation of the unit, including motor, switches, and all operating controls and mechanisms.

* 4.8 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The boring machine covered by this specification is intended for line boring and facing of engine block cam and main bearing bores during the rebuilding and reconditioning of internal combustion engines.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Measuring and indicating device graduation(s) required (see 3.2.2).
(d) Exceptions and additional safety/health requirements when required (see 3.4.8 and 6.4).
(e) Motor requirements if different (see 3.6.5.1).
(f) Power supply cord requirements if different (see 3.6.5.3).
(g) Standard equipment if different (see 3.8).
(h) Painting and finishing if different (see 3.9).
(i) Product identification if different (see 3.10).
(j) First article inspection if different (see 4.3).
(k) Quality conformance inspection if different (see 4.4).
(l) Sampling when required (see 4.5).
(m) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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